

Curriculum Vitae

Dr Michael Lydeamore
Department of Econometrics and Business Statistics
Monash University
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APPOINTMENTS HELD

Senior Lecturer - Econometrics and Business Statistics	2021-
Monash University	
COVID-19 Modelling & Forecasting Lead	2020
Manager, Analytics	
Department of Health and Human Services, Victoria	
Postdoctoral Research Fellow	2019-2021
Monash University	
Honorary Team Member	2019-
SaferCare Victoria	
Honorary Research Fellow	2019-
Alfred Health	
Research Fellow	2018-2019
The Kirby Institute, University of New South Wales	

EDUCATION

Doctor of Philosophy - Applied Mathematics	2015-2019
The University of Melbourne	
Thesis title: Mechanistic and statistical models of skin disease transmission	
Masters of Philosophy - Applied Mathematics	2013-2014
The University of Adelaide	
Thesis title: Approximations of stochastic household models for comparing antiviral allocation schemes	
Awarded the Applied Probability Trust prize for the highest mark with a project in Applied Mathematics and Statistics	
Bachelor of Mathematical Sciences	2010-2013
The University of Adelaide	
Graduated with a double major in applied and pure mathematics	

PUBLICATIONS

PREPRINTS

1. Gamage, J. P., Cook, D., Harrison, P., **Lydeamore, M.**, & Talagala, T. S. (2025). *quollr: An R Package for Visualizing 2-D Models from Nonlinear Dimension Reductions in High-Dimensional Space* (arXiv:2512.18166). arXiv. <https://doi.org/10.48550/arXiv.2512.18166>
2. Gamage, J. P., Cook, D., Harrison, P., **Lydeamore, M.**, & Talagala, T. S. (2025). *cardinalR: Generating Interesting High-Dimensional Data Structures* (arXiv:2512.18172). arXiv. <https://doi.org/10.48550/arXiv.2512.18172>

3. Gamage, J. P., Cook, D., Harrison, P., & **Lydeamore, M.** (2025). *Stop Lying to Me: New Visual Tools to Choose the Most Honest Nonlinear Dimension Reduction* (arXiv:2506.22051). arXiv. <https://doi.org/10.48550/arXiv.2506.22051>
4. Miller, C. M., **Lydeamore, M. J.**, Waddle, A. W., Berger, L., Skerratt, L. F., Flegg, J. A., & Campbell, P. T. (2025). *Mathematical modelling of chytridiomycosis transmission in frogs* (arXiv:2503.06846). arXiv. <https://doi.org/10.48550/arXiv.2503.06846>
5. Conway, E., Walker, C., **Lydeamore, M.**, Golding, N., Ryan, G., Mavec, D., Oates, J., Kabashima, G., Price, D. J., Shearer, F., Cromer, D., Davenport, M. P., McCaw, J., Eriksson, E. M., Hodgkin, P. D., Wu, L., Le, T. P., Baker, C. M., Mueller, I., & McVernon, J. (2024). *Optimal timing of booster doses in a highly vaccinated population with minimal natural exposure to COVID-19*. medRxiv. <https://doi.org/10.1101/2024.05.14.24307386>

PEER-REVIEWED ARTICLES

1. **Lydeamore, M.**, Zachreson, C., Conway, E., Shearer, F. M., Baker, C. M., Ross, J. V., Miller, J., McCaw, J. M., Geard, N. L., McVernon, J., & Price, D. J. (2026). Border quarantine, vaccination and public health measures to mitigate the impact of COVID-19 importations in Australia: a modelling study. *Journal of The Royal Society Interface*, 23(235), 20250144. <https://doi.org/10.1098/rsif.2025.0144>
2. Tierney, N., Saraswati, C., Babu, A., **Lydeamore, M.**, & Golding, N. (2026). conmat: generate synthetic contact matrices for a given age-stratified population. *Journal of Open Source Software*, 11(118), 8326. <https://doi.org/10.21105/joss.08326>
3. Nguyen, Q., **Lydeamore, M. J.**, Dessauer, A., Snell, G. I., Westall, G. P., Peleg, A. Y., & Gardiner, B. J. (2026). Evaluation of Global Immune Biomarkers to Predict Infections in Lung Transplant Recipients. *Clinical Transplantation*, 40(8), e70654. <https://doi.org/10.1111/ctr.70654>
4. Le, T. P., Conway, E., Akpan, E., Abell, I. R., Abraham, P., Baker, C. M., Campbell, P. T., Cromer, D., **Lydeamore, M. J.**, McDonough, Y., Mueller, I., Ryan, G., Walker, C., Wang, Y., Carvalho, N., & McVernon, J. (2025). Health impact and cost-effectiveness of COVID-19 booster vaccination strategies in the early post-Omicron era: a dynamic modelling study. *BMJ Global Health*, 10(9). <https://doi.org/10.1136/bmjgh-2024-016096>
5. Hao, T., Ryan, G. E., **Lydeamore, M. J.**, Cromer, D., Wood, J. G., McVernon, J., McCaw, J. M., Shearer, F. M., & Golding, N. (2025). Predicting immune protection against outcomes of infectious disease from population-level effectiveness data with application to COVID-19. *Vaccine*, 55, 126987. <https://doi.org/10.1016/j.vaccine.2025.126987>
6. **Lydeamore, M. J.**, Wu, D., Donker, T., Gorrie, C., Higgs, C. K., Easton, M., Hennessy, D., Geard, N., Howden, B. P., Cooper, B. S., Wilson, A., Peleg, A. Y., & Stewardson, A. J. (2024). Changes in isolation guidelines for CPE patients results in only mild reduction in required hospital beds. *Infection, Disease & Health*. <https://doi.org/10.1016/j.idh.2024.10.004>
7. **Lydeamore, M. J.**, Donker, T., Wu, D., Gorrie, C., Turner, A., Easton, M., Hennessy, D., Geard, N., Howden, B. P., Cooper, B. S., Wilson, A., Peleg, A. Y., & Stewardson, A. J. (2024). Carbapenemase-producing enterobacterales colonisation status does not lead to more frequent admissions: a linked patient study. *Antimicrobial Resistance & Infection Control*, 13(1), 82. <https://doi.org/10.1186/s13756-024-01437-x>
8. Shearer, F. M., Edwards, L., Kirk, M., Eales, O., Golding, N., Hassall, J., Liu, B., **Lydeamore, M.**, Miller, C., Moss, R., Price, D. J., Ryan, G. E., Sheena, S., Tobin, R., Ward, K., Kaldor, J., Cheng, A. C., Wood, J., & McCaw, J. M. (2024). Opportunities to strengthen respiratory virus surveillance systems in Australia: lessons learned from the COVID-19 response. *Communicable Diseases Intelligence*, 48. <https://doi.org/10.33321/cdi.2024.48.47>

9. Shearer, F. M., McCaw, J. M., Ryan, G. E., Hao, T., Tierney, N. J., **Lydeamore, M. J.**, Wu, L., Ward, K., Ellis, S., Wood, J., McVernon, J., & Golding, N. (2024). Estimating the impact of test–trace–isolate–quarantine systems on SARS-CoV-2 transmission in Australia. *Epidemics*, 47, 100764. <https://doi.org/10.1016/j.epidem.2024.100764>
10. Le, T. P., Abell, I., Conway, E., Campbell, P. T., Hogan, A. B., **Lydeamore, M. J.**, McVernon, J., Mueller, I., Walker, C. R., & Baker, C. M. (2024). Modelling the impact of hybrid immunity on future COVID-19 epidemic waves. *BMC Infectious Diseases*, 24(1), 407. <https://doi.org/10.1186/s12879-024-09282-4>
11. Conway, E., Walker, C. R., Baker, C., **Lydeamore, M. J.**, Ryan, G. E., Campbell, T., Miller, J. C., Rebuli, N., Yeung, M., Kabashima, G., Geard, N., Wood, J., McCaw, J. M., McVernon, J., Golding, N., Price, D. J., & Shearer, F. M. (2023). COVID-19 vaccine coverage targets to inform reopening plans in a low incidence setting. *Proceedings of the Royal Society B: Biological Sciences*, 290(2005), 20231437. <https://doi.org/10.1098/rspb.2023.1437>
12. Mitchell, B. G., Stewardson, A. J., Kerr, L., Ferguson, J. K., Curtis, S., Busija, L., **Lydeamore, M. J.**, Graham, K., & Russo, P. L. (2023). The incidence of nosocomial bloodstream infection and urinary tract infection in Australian hospitals before and during the COVID-19 pandemic: an interrupted time series study. *Antimicrobial Resistance & Infection Control*, 12(1), 61. <https://doi.org/10.1186/s13756-023-01268-2>
13. Zachreson, C., Shearer, F. M., Price, D. J., **Lydeamore, M. J.**, McVernon, J., McCaw, J., & Geard, N. (2022). COVID-19 in low-tolerance border quarantine systems: Impact of the Delta variant of SARS-CoV-2. *Science Advances*, 8(14), eabm3624. <https://doi.org/10.1126/sciadv.abm3624>
14. **Lydeamore, M. J.**, Mitchell, B. G., Bucknall, T., Cheng, A. C., Russo, P. L., & Stewardson, A. J. (2022). Burden of five healthcare associated infections in Australia. *Antimicrobial Resistance & Infection Control*, 11(1), 69. <https://doi.org/10.1186/s13756-022-01109-8>
15. Trauer, J. M., **Lydeamore, M. J.**, Dalton, G. W., Pilcher, D., Meehan, M. T., McBryde, E. S., Cheng, A. C., Sutton, B., & Ragonnet, R. (2021). Understanding how Victoria, Australia gained control of its second COVID-19 wave. *Nature Communications*, 12(1), 6266. <https://doi.org/10.1038/s41467-021-26558-4>
16. McMahon, J. H., **Lydeamore, M. J.**, & Stewardson, A. J. (2021). Bringing evidence from press release to the clinic in the era of COVID-19. *Journal of Antimicrobial Chemotherapy*, 76(3), 547–549. <https://doi.org/10.1093/jac/dkaa506>
17. Zachreson, C., Mitchell, L., **Lydeamore, M. J.**, Rebuli, N., Tomko, M., & Geard, N. (2021). Risk mapping for COVID-19 outbreaks in Australia using mobility data. *Journal of The Royal Society Interface*, 18(174), 20200657. <https://doi.org/10.1098/rsif.2020.0657>
18. Sullivan, S. G., Brotherton, J. M., Lynch, B. M., Cheung, A., **Lydeamore, M.**, Stevenson, M., Firestone, S., Canevari, J., Nguyen, H. N. J., & Carville, K. S. (2021). Population-based analysis of the epidemiological features of COVID-19 epidemics in Victoria, Australia, January 2020–March 2021, and their suppression through comprehensive control strategies. *Lancet Regional Health - Western Pacific*, 17.
19. **Lydeamore, M. J.** (2021). Mathematical models to support Victoria’s COVID-19 response: a blunt instrument to a complex problem. *Journal of the Australian Mathematical Society*. <https://austms.org.au/wp-content/uploads/2021/07/Lydeamore.pdf>
20. **Lydeamore, M. J.**, Campbell, P. T., Price, D. J., Wu, Y., Marcato, A. J., Cuningham, W., Carapetis, J. R., Andrews, R. M., McDonald, M. I., McVernon, J., Tong, S. Y. C., & McCaw, J. M. (2020). Estimation of the force of infection and infectious period of skin sores in remote Australian communities using interval-censored data. *PLOS Computational Biology*, 16(10), e1007838. <https://doi.org/10.1371/journal.pcbi.1007838>

SOFTWARE

R PACKAGES

blockstrap

CRAN

Sample dataframes by group, in the form of a 'block bootstrap'. Entire groups are returned allowing for a single 'observation' to span multiple rows of the dataframe.

Coauthors: Michael Lydeamore, Cash Looi, Kenyon Ng, Mitchell O'Hara-Wild

cardinalR

CRAN

A collection of functions to generate a large variety of structures in high dimensions. These data structures are useful for testing, validating, and improving algorithms used in dimensionality reduction, clustering, machine learning, and visualization.

Coauthors: Jayani P. Gamage, Dianne Cook, Paul Harrison, Michael Lydeamore, Thiyanga S. Talagala

condensr

CRAN

Helps automate 'Quarto' website creation for small academic groups.

Builds a database-like structure of people, projects and publications, linking them together with a string-based ID system. Then, provides functions to automate production of clean markdown for these structures, and in-built CSS formatting using CSS flexbox.

Coauthors: Michael Lydeamore

HospitalNetwork

CRAN

Set of tools to help interested researchers to build hospital networks from data on hospitalized patients transferred between hospitals. Methods provided have been used in Donker T, Wallinga J, Grundmann H. (2010) <doi:10.1371/journal.pcbi.1000715>, and Nekkab N, Crépey P, Astagneau P, Opatowski L, Temime L. (2020) <doi:10.1038/s41598-020-71212-6>.

Coauthors: Pascal Crépey, Tjibbe Donker, Clément Massonnaud, Michael Lydeamore

mapycusmaximus

CRAN

Focus-glue-context (FGC) fisheye transformations to two-dimensional coordinates and spatial vector geometries. Implements a smooth radial distortion that enlarges a focal region, transitions through a glue ring, and preserves outside context. Methods build on generalized fisheye views and focus+context mapping. For more details see Furnas (1986) <doi:10.1145/22339.22342>, Furnas (2006) <doi:10.1145/1124772.1124921> and Yamamoto et al. (2009) <doi:10.1145/1653771.1653788>.

Coauthors: Alex Nguyen, Dianne Cook, Michael Lydeamore

quollr

CRAN

To construct a model in 2-D space from 2-D nonlinear dimension reduction data and then lift it to the high-dimensional space. Additionally, provides tools to visualise the model overlay the data in 2-D and high-dimensional space. Furthermore, provides summaries and diagnostics to evaluate the nonlinear dimension reduction layout.

Coauthors: Jayani P. Gamage, Dianne Cook, Paul Harrison, Michael Lydeamore, Thiyanga S. Talagala

VS CODE EXTENSIONS

Code Comment Viewer

[Open VSX](#)

Show all HTML comments in the current file in a sidebar and highlight them in the editor.

Version 0.1.3

Collab Comments

[Open VSX](#)

Track comments with author and timestamp for QMD and other files

Version 1.1.0

Comment Tracker

[Open VSX](#)

Track comments with author and timestamp for QMD and other files

Version 1.0.0

Quarto: Open Rendered File

[Open VSX](#)

Open an existing rendered output for a Quarto document without rendering it again.

Version 1.1.2

FUNDING

Almost all of this funding was obtained in partnership with large, collaborative teams.

SPARKLE

2024

Department of Foreign Affairs and Trade

Awarded value: \$5,000,000

SHIELD: Surveillance of Healthcare-associated Infections for Effective Local Data

2025

NHMRC

Awarded value: \$2,000,000

Modelling Work in Low and Middle Income Counties in the Western Pacific Region for the COVID-19 SAGE Working Group

2024

World Health Organisation

Awarded value: \$8,157

Modelling To Support Australia's Resilience to and Preparedness for Omicron And Future Sars-Cov-2 Variants

2024

Commonwealth Government of Australia

Awarded value: \$80,059

Contractual Partner to Conduct Research on Healthcare-Associated Infection Rates in Phillippine Hospitals

2023

World Health Organisation Phillippines

Awarded value: \$72,078

Centre of Western Public Health Unit

2023

Contract Research

Awarded value: \$25,000

Extending and comparing methods for projecting social contact matrices

2022

SPECTRUM/SPARK Seed Funding

Awarded value: \$19,438

Quantifying longitudinal relationships between community mobility and COVID-19 case incidence in west metropolitan Melbourne

2022

SPECTRUM/SPARK Seed Funding

Awarded value: \$19,896

Modelling to support Australia's national plan for COVID-19	2021
Commonwealth Government of Australia	
Awarded value: \$147,536	
Modelling to support Australia's transition to 'COVID-normal'	2021
Australian Office of Health Protection	
Awarded value: \$25,620	

CONFERENCE PRESENTATIONS AND CONTRIBUTED TALKS

Gamifying data visualisation: Teaching ggplot2 through competitive code golf	2026
International Congress on Teaching Statistics	
Gamifying data visualisation: Teaching ggplot2 through competitive code golf	2026
useR! Conference	
Gamifying data visualisation: Teaching ggplot2 through competitive code golf	2026
Australia and New Zealand Industrial Applied Mathematics Conference	
Modelling chytridiomycosis transmission in frogs	2025
Australia and New Zealand Industrial Applied Mathematics Conference	
airpurifyr: Open Air Quality Data in R	2024
WOMBAT Annual Meeting: Open the world with open source	
Networks of networks in infectious diseases modelling - Panelist	2024
Infectious Diseases Modelling Conference	
Generating synthetic contact matrices using open-source data	2024
Australia and New Zealand Industrial Applied Mathematics Conference	
Data-Driven Insights into Healthcare Challenges: Two Case Studies - Invited Speaker	2023
Universitätsklinikum Freiburg Seminar Series	
Data-Driven Insights into Healthcare Challenges: Two Case Studies	2023
UNSW Australia Statistics & Data Science Seminar	
Burden of healthcare associated infections in Australia	2022
Australia and New Zealand Industrial Applied Mathematics Conference	
Exponential Random Graph Models and CPE transmission - Invited Speaker	2021
Melbourne Mathematical Biology Seminar Series	
Mathematical modelling for COVID-19 in Victoria, Australia - Invited Speaker	2021
Math for Industry Forum, Vietnam	
The associations between transmission of CPE and ward connectivity: a network analysis	2021
Australasian Society for Infectious Diseases Annual Scientific Meeting	
The burden of healthcare acquired infections in Australian public hospitals	2021
Australasian Society for Infectious Diseases Annual Scientific Meeting	
Decision Making and Mathematical Biology - Victoria's usage of COVID-19 modelling -	2021
Invited Speaker	
Mathematical Biology Special Interest Group Workshop	
Modelling between-household effective contact and the elimination of COVID-19 in Melbourne, Australia	2021
Australia and New Zealand Industrial Applied Mathematics Conference	
Mathematical modelling and Victoria's response to COVID-19 - Invited Speaker	2020
Victorian ANZIAM Branch Meeting	

Australia's experience and the role of modelling in its responses to COVID-19	- Invited	2020
Speaker		
Usher Institute COVID-19 Webinar		
Estimating epidemiological quantities for skin sores in remote Australian communities using interval-censored data		2019
Australia and New Zealand Industrial Applied Mathematics Conference		
Estimating epidemiological quantities for skin sores in remote Australian communities using interval-censored data		2018
NSW-ACT ANZIAM Branch Meeting		
Coupled models of Group A Streptococcus and Scabies: How likely is eradication?		2018
PRISM International Conference		
Investigating the dynamics of coupled epidemiological transmission models with application to Group A Streptococcus and Scabies		2018
Society for Mathematical Biology Annual Meeting		
Investigating the dynamics of coupled models with applications to Group A Streptococcus and Scabies		2018
Australia and New Zealand Industrial Applied Mathematics Conference		
Quantifying the age of first infection with skin sores in five remote Australian Aboriginal communities		2017
Lancefield International Symposium on Streptococci and Streptococcal Diseases		
Investigating the dynamics of coupled models with applications to Group A Streptococcus and Scabies		2017
PRISM Annual Knowledge Transfer and Training Conference		
Constructing mathematical models of Group A Streptococcus and Scabies in remote Australian Indigenous communities	- Invited Speaker	2017
Zeeman Institute Seminar Series		
Developing a model for the transmission and treatment dynamics of scabies infections at high prevalence setting		2017
Melbourne-Manchester Joint Workshop		
Developing a model for the transmission of Group A Streptococcus		2017
Australia and New Zealand Industrial Applied Mathematics Conference		
Determining the age of first infection from incomplete data		2016
Modelling Emerging Infections and Neglected Tropical Diseases Workshop		
Investigating Intervention Intervals for Scabies Infections		2016
Australia and New Zealand Industrial Applied Mathematics Conference		

TEACHING

LECTURER

[ETC5513] Reproducible and Collaborative practices	- Chief Examiner	2024-
[ETC5523] Communicating with Data	- Chief Examiner	2023-
[ETO5513] Reproducible and Collaborative practices	- Chief Examiner	2024-
[ETC5512] Wild-Caught Data		2022-2023
[ETC5521] Exploratory Data Analysis		2022

TUTOR

[MAST10016] Mathematics for Biomedicine		2016-2019
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[MAST30001] Stochastic Modelling	2018
TEACHING SERVICE	
[M6036] Masters of Health Data Analytics - Course Management Committee	2024-
AWARDS	
Fellow of the Higher Education Academy	2024
SPECTRUM Annual Meeting - Best Presentation	2024
University of Melbourne Faculty of Engineering and Information Technology Excellence Award in Interdisciplinary Research	2021
IPAA Spirit of Service Awards - Finalist	2021
Engagement Australia Excellence Awards - Outstanding Engagement for Research Impact - Finalist	2021
Top Poster Award - European Congress of Clinical Microbiology & Infectious Diseases	2021
IPAA Victoria Leadership in the Public Sector Awards - Finalist	2021
Best PhD Student Presentation - PRISM² Annual Conference	2017
Applied Probability Trust Prize - for best Applied Mathematics Postgraduate Thesis	2016
Australian Postgraduate Award	2015
Dean's Commendation for Thesis Excellence	2015
ACADEMIC SERVICE & COMMUNITY ENGAGEMENT	
Equity Diversity & Social Inclusion Committee Department representative	2024-
International Conference on Prevention and Infection Control Abstract Reviewer	2025
SPECTRUM Annual Meeting Organising committee	2024
SPARK Short Course in Mathematical Diseases Modelling - OUCRU Facilitator	2023
International Conference on Prevention and Infection Control Abstract Reviewer	2023
Maths in Industry Study Group - NSW Health Moderator	2023
WOMBAT Communicating with Data Workshop Organising Committee	2023
Australian and New Zealand Industrial and Applied Mathematics Executive Committee Treasurer	2022-
Australian and New Zealand Industrial and Applied Mathematics Executive Committee Early Career Representative	2021-2022
SPECTRUM-SPARK Early Career Researcher Committee Chair	2022-
SPARK Short Course in Mathematical Diseases Modelling - Mahidol University Facilitator	2022
Research Tools Workshop in R Facilitator	2022

Mathematical Biology Special Interest Group	<i>2019-2022</i>
Treasurer	
ANZIAM 2021 Conference	<i>2020-2021</i>
Organising Committee (Treasurer)	
Computational Biology Research Initiative - The University of Melbourne	<i>2017</i>
Postgraduate Representative	
Computational Biology Postgraduate Sports Group - The University of Melbourne	<i>2017-2018</i>
Secretary	
Maths in Industry Study Group	<i>2017, 2018, 2019</i>
Participant	
The ConocoPhillips Science Experience - The University of Melbourne	<i>2016-2018</i>
Activity Organiser	
Mathematicians in Schools - CSIRO	<i>2016-2018</i>
Project Supervisor/Mentor	
Reviewer	
PLOS Computational Biology; Scientific Reports; Journal of Applied Mathematics; Journal of Statistical Software; Epidemiology & Infection; Mathematics; Medical Journal of Australia; International Tropical Health; Journal of Antimicrobial Resistance; The R Journal	

PROFESSIONAL MEMBERSHIPS

Australian and New Zealand Industrial and Applied Mathematics	<i>2015-</i>
Including the Mathematical Biology Special Interest Group	
Australian Mathematics Society	<i>2015-</i>
Statistical Society of Australia	<i>2024-</i>